

Empirical Task Methodology & Continuous Injection Mapping

Formal Protocol Definition for 2AFC Probabilistic Reversal Data Ingestion

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August 15, 2026

Abstract

This technical methodology report defines the exact data dictionary, temporal structure, and discrete-to-continuous injection mapping for the 2AFC Probabilistic Reversal behavioral dataset (`behavioral_compile.csv`). Comprising 16,029 trials across 128 human participants, the task pairs explicit reward magnitudes ($M_A, M_B \in \{2..8\}$) with hidden probabilistic reward contingencies (85/15 vs 15/85). We formulate a mathematically rigorous continuous-time Mossy Fiber injection bridge ($\mathbf{u}_t$) with $\Delta t = 10$ ms resolution, enabling seamless deployment of the `ExactRModel` continuous-time C++ cerebellar reservoir.

1 Behavioral Task Design & Experimental Structure

The empirical dataset captures human decision-making in a 2-Alternative Forced Choice (2AFC) Probabilistic Reversal Learning paradigm:

1. **Stimulus Presentation:** Participants are presented with two options (A and B) displaying explicit numerical reward magnitudes ($M_A = \text{Bd1}, M_B = \text{Bd2} \in \{2, 3, 4, 5, 6, 7, 8\}$).
2. **Hidden Probabilistic Contingency:** One option offers a high reward probability (85%), while the other offers a low reward probability (15%). Contingency states periodically reverse without warning.
3. **Stochastic Feedback:** Upon choice selection, participants receive either the full magnitude (M_A or M_B) if rewarded ($F = 1$), or 0 points if unrewarded ($F = 0$).

2 Data Dictionary & Summary Statistics

The raw dataset `behavioral_compile.csv` contains 16,029 rows across 128 subjects.

Table 1: Empirical Dataset Data Dictionary (`behavioral_compile.csv`).

Variable	Type	Description & Range
<code>participant_id</code>	String	Unique subject identifier (e.g., <code>ACM0_06011994</code>).
<code>nt</code>	Integer	Sequential trial index within subject (1...130).
<code>buena</code>	Binary	Latent state (1 $\implies$ <i>A</i> is 85% optimal; 0 $\implies$ <i>B</i> is 85% optimal).
<code>Bd1 (M_A)</code>	Integer	Explicit reward magnitude for Option A (2...8).
<code>Bd2 (M_B)</code>	Integer	Explicit reward magnitude for Option B (2...8).
<code>Resp</code>	Categorical	Choice executed (1 $\implies$ Option A; 2 $\implies$ Option B).
<code>F</code>	Binary	Feedback outcome (1 $\implies$ Reward Delivered; 0 $\implies$ Zero Points).
<code>prob</code>	Continuous	True underlying probability for Option A (0.15 or 0.85).
<code>ttp</code>	Float (ms)	Timestamp of Stimulus Presentation onset.
<code>ttr</code>	Float (ms)	Timestamp of Participant Response execution.
<code>ttF</code>	Float (ms)	Timestamp of Feedback Presentation onset.

2.1 Summary Statistics

- **Total Participants:** 128
- **Total Observations:** 16,029 trials (Mean 125.2 trials/subject)
- **Overall Accuracy:** 49.86% choice of optimal contingency option
- **Overall Reward Rate:** 63.42%

3 The Continuous Injection Bridge ($\mathbf{u}_t$ Mapping)

Because the C++ `ExactRModel` cerebellar reservoir operates in continuous time ($\Delta t = 10$ ms), discrete event timestamps (`ttp` $\rightarrow$ `ttr` $\rightarrow$ `ttF`) must be transformed into a 5-dimensional Mossy Fiber input vector $\mathbf{u}(t) \in \mathbb{R}^5$:

$$\mathbf{u}(t) = [M_A(t), M_B(t), \text{Choice}_A(t), \text{Choice}_B(t), \text{Outcome}(t)]^T \quad (1)$$

3.1 Temporal Phase Array Padding Scheme

For each trial k , the discrete interval $[t_0, t_f]$ is discretized into time steps $t = m \cdot \Delta t$:

1. **Stimulus Window** ($t \in [\text{ttp}_k, \text{ttr}_k]$):

$$M_A(t) = \text{Bd1}_k, \quad M_B(t) = \text{Bd2}_k, \quad \text{Choice}_{A,B}(t) = 0, \quad \text{Outcome}(t) = 0$$

2. **Response Window** ($t \in [\text{ttr}_k, \text{ttF}_k]$):

$$\text{Choice}_A(t) = \mathbb{I}(\text{Resp}_k = 1), \quad \text{Choice}_B(t) = \mathbb{I}(\text{Resp}_k = 2), \quad \text{Outcome}(t) = 0$$

3. **Feedback Window** ($t \in [\text{ttF}_k, \text{ttF}_k + 200\text{ms}]$):

$$\text{Outcome}(t) = F_k \cdot M_{\text{chosen},k}$$

4. **Inter-Trial Interval (ITI)** ($t \in [\text{ttF}_k + 200\text{ms}, \text{ttp}_{k+1}]$):

$$\mathbf{u}(t) = \mathbf{0}$$

This continuous bridge preserves exact sensorimotor dynamics, ensuring the Granule cell layer receives magnitude stimuli prior to decision execution.